Phase-22 Uncertainty Bands (r3)

- Uncertainty CSV: `outputs/phase22/p22_uncertainty.csv`
- Bands JSON: `outputs/phase22/p22_bands.json`
- Rows: **50**
- Targets: `[0.9, 0.94, 0.96, 0.98]`
- grid: `[2.7, 2.721, 2.742, 2.763, 2.783, 2.804, 2.82, 2.825, 2.846, 2.867, 2.888, 2.9, 2.908, 2.929, 2.94, 2.95]` (|grid|=16)

Band width summary

- band_width: n=50, mean=1.066, median=1.261, p05=0.06504, p95=1.437
- band_width_CI_lo: n=50, mean=0.6123, median=0.6603, p05=0, p95=1.297
- band_width_CI_hi: n=50, mean=4.517, median=1.588, p05=0.1447, p95=2.824

Example rows

target		K_lo	K_hi	width	(K)	s_i(mid)
0.900	2.700	0.188	1.600	1.412	0.030	0.963
0.940	2.700	0.315	1.600	1.285	0.393	0.968
0.960	2.700	0.506	0.595	0.088	0.114	0.958
0.900	2.721	0.172	1.600	1.428	0.864	0.974
0.940	2.721	0.293	1.600	1.307	0.004	0.976
0.960	2.721	0.438	1.562	1.123	0.392	0.977
0.980	2.721	0.575	1.425	0.850	0.392	0.977
0.900	2.742	0.163	1.600	1.437	0.081	0.984
0.940	2.742	0.269	1.600	1.331	0.045	0.985
0.960	2.742	0.459	1.541	1.083	0.178	0.985

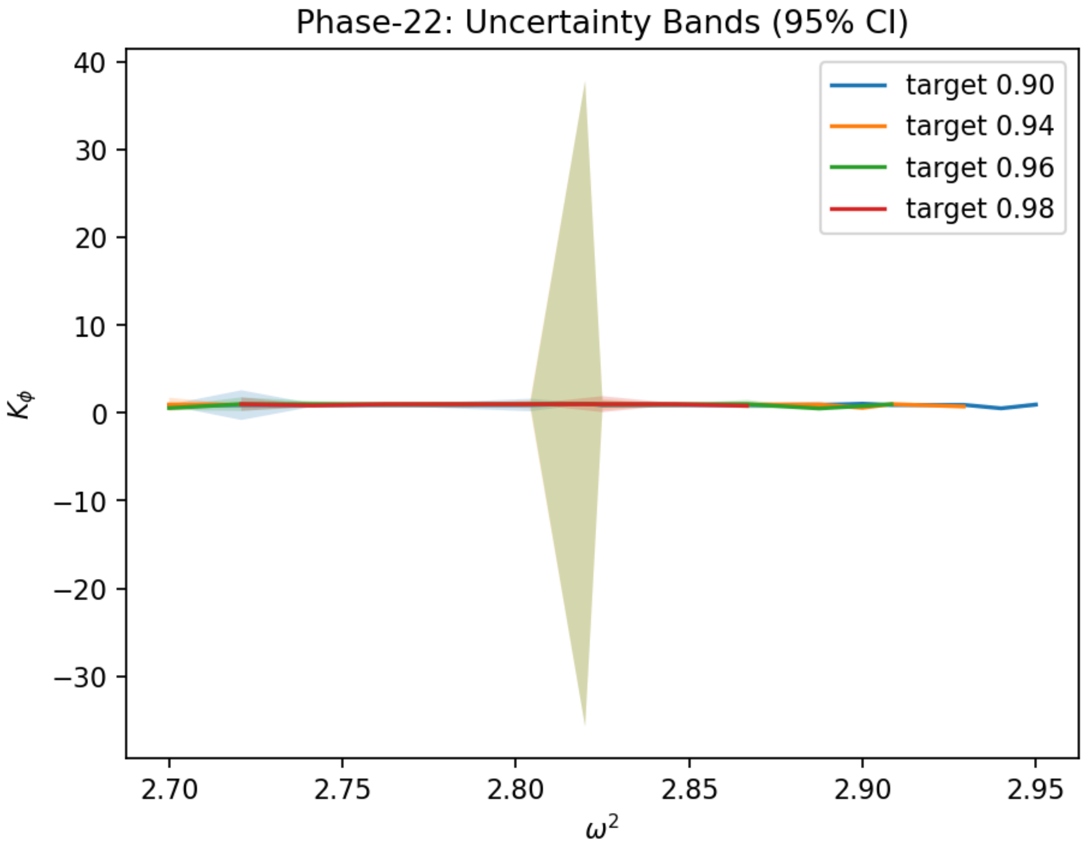
Coverage grid (band exists, none)

target \	2.70	2.72	2.74	2.76	2.78	2.80	2.82	2.83	2.85	2.87	2.89	2.90	2.91	2.93	2.94	2.95
0.90																
0.94																
0.96																
0.98																

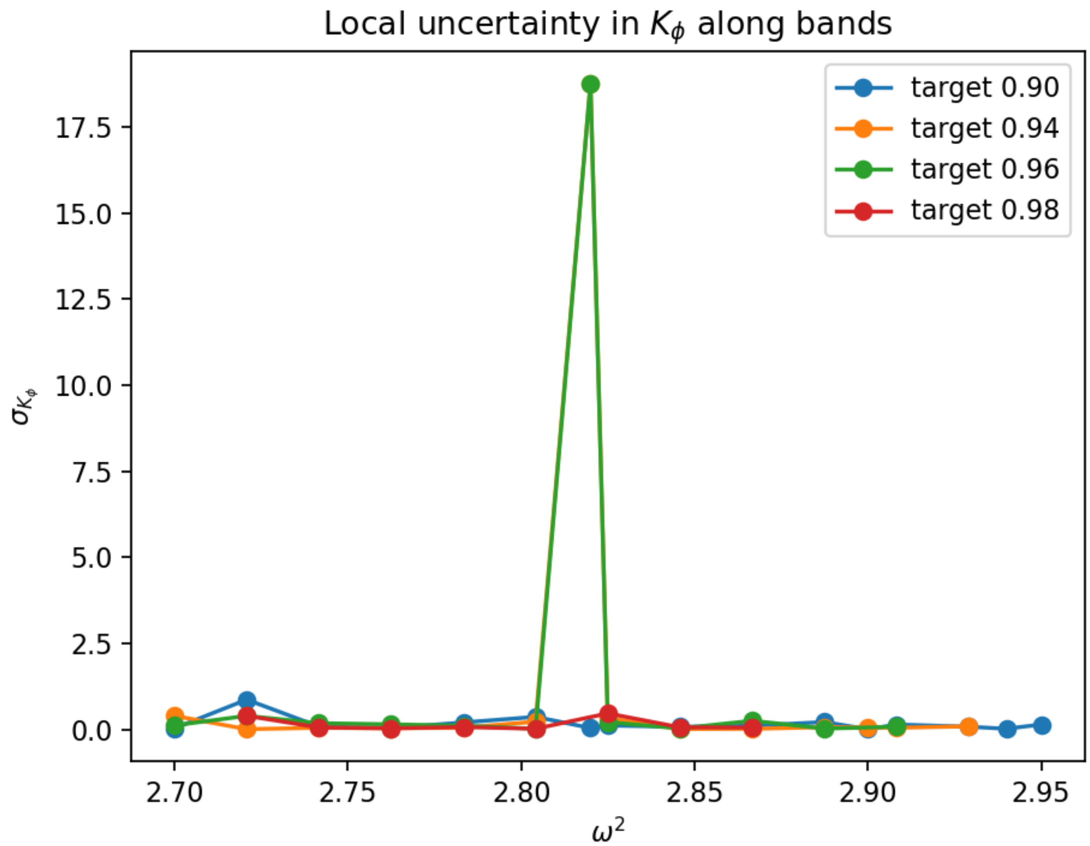
Method

We estimate the local slope at the midpoint as $(\partial s_i/\partial K_\phi)$, bootstrap s_i, and propagate to σ_{K_ϕ} via linear error propagation. Flat slopes inflate σ_{K_ϕ} by design.

Bands (with CIs)



(K) map


Local uncertainty in K_ϕ along bands
target 0.90
target 0.94
target 0.96
target 0.98
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
σ_{K_ϕ}
2.70
2.75
2.80
2.85
2.90
2.95
ω^2